Szamuely Chapter 1

Kyle Binder

September 10, 2021

1 Infinite Galois Theory

Let us begin be looking at the Galois theory for infinite field extensions. Having this as a background will help us to understand Grothendieck's formulation better.

Let K/k be a Galois extension (this means that, if G is the Galois group, $K^G = k$). Form an inverse system of finite groups consisting of the Galois groups of *finite* Galois subextension of k contained in K. Thus for $K \supset E \supset k$ Galois, we have $G_E = \mathrm{Gal}(E/k)$. Moreover, for all $E \supset F \supset k$ we have a surjective map (from the Fundamental Theorem of Galois Theory) $\varphi_{EF} : G_E \to G_F$. In essence this is the restriction map taking a k-automorphism in E and restricting to F. This may also be viewed as a quotient map by $\mathrm{Gal}(E/F)$.

This inverse system has inverse limit which is isomorphic to $\mathrm{Gal}(K/k)$. Therefore we may put a topology on this Galois group by placing the discrete topology on each G_E, and viewing $\mathrm{Gal}(K/k)$ as a subspace of $\prod_{E \text{ Galois}} G_E$.

With this topology, the Fundamental Theorem of (Infinite) Galois Theory reads:

Theorem 1. *Let L be a subextension of the Galois extension K/k. Then $\mathrm{Gal}(K/L)$ is a closed subgroup of $\mathrm{Gal}(K/k)$. Moreover, the maps*

$$L \mapsto \mathrm{Gal}(K/L) \quad \textit{and} \quad H \mapsto K^H$$

give an inclusion reversing bijection between subfields $K \supset L \supset k$ and closed subgroups $H \subset G$. A subextension L/k is Galois over k if and only if $\mathrm{Gal}(K/L)$ is normal in $\mathrm{Gal}(K/k)$. In this case there is a natural isomorphism $\mathrm{Gal}(L/k) \cong \mathrm{Gal}(K/k)/\mathrm{Gal}(K/L)$.

2 Category Theory

We need one more tool before we are able to reach Grothendieck's reformulation of Galois theory, and that is *representable functors*.

Here I assume knowledge of the definitions of *category*, *functor*, the *product category*, (*full*) *subcategory*, *full* and *faithful functors*, and *essentially surjective*.

Definition 1. A *natural transformation* (called a *morphism of functors* in Szamuely) between two functors $F, G : \mathcal{C}_1 \to \mathcal{C}_2$, is a family of maps $\varphi_A : F(A) \to G(A)$ indexed over the objects of $\mathcal{C}_1$ making the following diagram commute for all $\varphi \in \mathrm{Hom}(A, B)$.

$$\begin{tikzcd} F(A) \arrow[r, "\varphi_A"] \arrow[d, "F(\varphi)"] & G(A) \arrow[d, "G(\varphi)"] \\ F(B) \arrow[r, "\varphi_B"] & G(B) \end{tikzcd}$$

With the defintion of natural transformations, we may describe *equivalent*, or *quasi-isomorphic*, categories.

Definition 2. Two categories $\mathcal{C}_1 \to \mathcal{C}_2$ are *equivalent* if there are two covariant functors $F, G : \mathcal{C}_1 \to \mathcal{C}_2$ with $F \circ G$ naturally isomorphic to $\mathrm{id}_{\mathcal{C}_2}$ and $G \circ F$ naturally isomorphic to $\mathrm{id}_{\mathcal{C}_1}$.

If the functors are instead contravariant, we say the categories are *anti-equivalent*

Another definition of equivalence of categories is found in the following lemma, which I assume most have seen. The proof is more or less following your nose.

Lemma 1. *Two categories $\mathcal{C}_1, \mathcal{C}_2$ are equivalent if and only there is a covariant functor $F : \mathcal{C}_1 \to \mathcal{C}_2$ which is fully faithful and essentially surjective.*

An important class of functors are the *representable functors*. These always take a category to the category of sets.

Definition 3. Let $\mathcal{C}$ be a category. A functor $F : \mathcal{C} \to \underline{Set}$ is *representable* if there is an object $C \in \mathcal{C}$ and a natural isomorphism $F \cong \mathrm{Hom}(C, \cdot)$.

In this case, C is a called an *representing object*.

For representable functors we have the ubiquitous Yoneda Lemma.

Lemma 2 (Yoneda)**.** *Let $F, G : \mathcal{C} \to \underline{Set}$ be representable functors with objects C, D respectively. Then every natural transformation is give by a unique $\varphi : D \to C$.*

As a corollary, we have the famous tagline "an object is determined by its morphisms."

Corollary 1. *The representing object of a representable functor F is unique up to unique isomorphism.*

3 Finite Etale Algebras

We may now describe Grothendieck's reformulation of Galois theory.

Take a base field k and fix a separable closure k_s and algebraic closure $\bar{k}$. We will abbreviate $\mathrm{Gal}(k_s/k)$ as $\mathrm{Gal}(k)$. For L/k separable (though not necessarily in k_s) ther are $[L : k]$ k-algebra homomorphisms $L \hookrightarrow \bar{k}$, and the images of these are in k_s. Therefore $\mathrm{Hom}_k(L, k_s)$ is finite, and there is a natural left $\mathrm{Gal}(k)$-action which is just post-composition.

Lemma 3. *The action of $\mathrm{Gal}(k)$ on $\mathrm{Hom}_k(L, k_s)$ is continuous and transitive. Therefore $\mathrm{Hom}_k(L, k_s)$ is isomorphic over $\mathrm{Gal}(k)$ to the left coset space of some open subgroup in $\mathrm{Gal}(k)$. For L Galois over k this coset space is a quotient by an open normal subgroup.*

Proof. The stabilizer G_φ are the elements of $\mathrm{Gal}(k)$ which fix $\varphi(L)$. This is a closed subset by the fundamental theorem, and has finite index (the cosets are bounded by $|\mathrm{Gal}(L/k)|$), so is therefore open. This implies the action is continuous.

To see the action is transitive, write $L = K(\alpha)$, and then note any $\varphi \in \mathrm{Hom}_k(L.k_s)$ is determined by $\varphi(\alpha)$ which must get mapped to one of the roots of the minimal polynomial of α. Moreover, as $\mathrm{Gal}(k)$ acts transitively on these roots, we have the $\mathrm{Gal}(k)$-action on $\mathrm{Hom}_k(L, k_s)$ is transitive. Therefore $\mathrm{Hom}_k(L, k_s)$ is isomorphic to the left coset space $G_\varphi \backslash \mathrm{Gal}(k)$.

In the case G_φ is normal, this is $\mathrm{Gal}(k)/G_\varphi$, and this occurs if and only if L/k is Galois. $\square$

Now take M a finite separable extension of k, then by precomposition, each k-homomorphism $\varphi : L \to M$ gives a map $\mathrm{Hom}_k(M, k_s) \to \mathrm{Hom}_k(L, k_s)$, and yields a contravariant functor from the category of finite separable extensions of k to the category of finite sets with continuous transitive left $\mathrm{Gal}(k)$-action.

This functor turns out to be an anti-equivalence of categories.

Theorem 2. *Let k be a field with fixed separable closure k_s. The contravariant functor mapping a finite separable extension L/k to the finite $\mathrm{Gal}(k)$-set $\mathrm{Hom}_k(L, k_s)$ gives an anti-equivalence between finite separable extensions of k and finite sets with continuous and transitive left $\mathrm{Gal}(k)$-actions. Here Galois extensions give rise to $\mathrm{Gal}(k)$-sets isomorphic to some finite quotient of $\mathrm{Gal}(k)$.*

Proof. We begin with essential surjectivity. Take S a $\mathrm{Gal}(k)$-set. Fix $s \in S$; the stabilizer G_s is an open subgroup which fixes the finite separable extension L/s. Define $\mathrm{Hom}_k(L, k_s) \to S$ by $g \circ i \mapsto gs$ where i is the natural inclusion $L \to k$. This map is an isomorphism.

We now show the map is full faithful. Take L, M finite separable extensions of k. As $\mathrm{Hom}_k(M, k_s)$ and $\mathrm{Hom}_k(L, k_s)$ are transitive $\mathrm{Gal}(k)$-sets, a $\mathrm{Gal}(k)$-equivariant map f between them is determined by the image of one fixed $\varphi \in \mathrm{Hom}_k(M, k_s)$. Now $G_\varphi \subset G_{f(\varphi)}$ by equivariance. Therefore $f(\varphi)(L) \subset \varphi(M)$. Now let $\psi : \varphi(M) \to M$ be the inverse of φ. Then $\psi \circ f(\varphi)$ is the unique element of $\mathrm{Hom}_k(L, M)$ fixing f.

The last statement follows from the previous lemma. □

The only limitation now before we apply this to get an equivalence of $\mathrm{Gal}(k)$-sets is that the action on a general set may not be transitive. However, a finite separable extension will always have transitive Galois action. To get around this, we will need to consider étale k-algebras.

Definition 4. A finite dimensional k-algebra A is *étale* over k if it is isomorphic to a finite direct product of (finite) separable extensions of k.

Theorem 3 (Main Theorem of Galois Theory – Grothendieck)**.** *Let k be a field. The functor mapping a finite étale k-algebra A to the finite set $\mathrm{Hom}_k(A, k_s)$ gives an anti-equivalence between the category of finite étale k-algebras and the category of finite sets with continuous left $\mathrm{Gal}(k)$-actions. Here separable field extensions give rise to sets with transitive actions, and Galois extensions give rise to $\mathrm{Gal}(k)$-sets isomorphic to finite quotients of $\mathrm{Gal}(k)$.*

Proof. Take a decomposition $A = \oplus L_i$ into seprable extensions. Then a non-constant $\varphi \in \mathrm{Hom}_k(A, k_s)$ induces and embedding of exactly one of the L_i in k_s. Indeed, take an element of a L_i not mapping to 0. Then $L_i \hookrightarrow k_s$. But if another field injected into k_s as well there would be zero-divisors.

Therefore

$$\mathrm{Hom}_k(A, k_s) \cong \coprod \mathrm{Hom}_k(L_i, k_s)$$

Using the previous result this gives a decomposition into Galois orbits of a Gal_k set. □

To conclude this talk, I will give some equivalent formulations for a finite dimensional commutative k-algebra to be étale.

Proposition 1. *Let A be a finite dimensional commutative k-algebra. Then the following are equivalent:*

1. *A is étale.*
2. *$A \otimes_k \bar{k}$ is isomorphic to a finite product of copies of $\bar{k}$.*
3. *$A \otimes \bar{k}$ is reduced.*

Before we prove this theorem we first need a lemma.

Lemma 4. *A finite dimensional commutative algebra over a field F is isomorphic to a direct product of finite field extensions of F if and only if it is reduced.*

Proof. The only if part of this result is immediate. For the other direction, by writing the F-algebra A as a finite direct product of indecomposables, we may assume that A is indecomposable. Notice that A may only have 1 and 0 as idempotents or else we may further decompose A.

We will show that A is a field. Take $x \neq 0$ in A. As A is finite dimensional, $(x) \supset (x^2) \supset \cdots$ must stabilize, so for some n and y we have

$$x^n = x^{n+1}y = x^{n+2}y^2 = \cdots = x^{n+i}y^i$$

for all $i \in \mathbb{N}$. This says $x^n = x^{2n}y^n$, and therefore $x^n y^n = (x^n y^n)^2$. Note neither x or y may be zero, as $x \neq 0$ and $x^n x^{2n} y^n$ and A reduced. Therefore $1 = x^n y^n$ and x is invertible. $\square$

We are now in a position to prove the lemma.

Proof. The previous lemma implies that the last two statements are equivalent.

To see the first implies the second, we just need to show that $L \otimes_k \bar{k}$ is a finite product of $\bar{k}$ when L/k is finite and separable. Indeed,

$$L \otimes_k \bar{k} \cong \bar{k}[\alpha] \cong \bar{k}[x]/(f_\alpha) \cong \prod \bar{k}[x]/(x - \alpha_i) \cong \prod \bar{k}$$

where α_i are the roots of f_α the minimal polynomial of α over k.

To see (2) implies (1), take $\overline{A}$ to be the quotient of A by its nilpotents. Therefore $\overline{A}$ is reduced, and $\overline{A}$ is a sum of finite extension fields of k. As $\bar{k}$ is reduced, each k-algebra homomorphism $A \to \bar{k}$ factors through $\overline{A}$ and therefore through one of its separable field factors L. The number of such homomorphisms is bounded by $[L : k]$, with equality if and only if L is separable. Thus $\mathrm{Hom}_k(A, \bar{k})$ has at most $\dim_k(A)$ elements with equality if and only if $A = \bar{A}$ and A is étale.

This equality holds as

$$\mathrm{Hom}_k(A, \bar{k}) \cong \mathrm{Hom}_{\bar{k}}(A \otimes_k \bar{k}, \bar{k})$$

$\square$